Adaptive Reasoning: An Affect Related Feedback Approach for Enhanced E-learning

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Abstract. Recognition of affective states to enhance e-learning platforms has been a topic of machine learning research. Compared to other input modalities, facial expressions have the potential to reveal nonverbal cues about a learner's learning affect. However, most studies were limited in their analysis of learning affects exhibited by a learner with the possibility of providing appropriate feedback to teachers and learners. This work proposes an adaptive reasoning mechanism that considers the estimated affective states and learning affect in generating feedback with reasoning incorporated. This work utilizes a Convolutional Neural Network-Bidirectional Long-Short Term Memory (CNN-BiLSTM) cascade framework for affective states analysis through processing a live/stored observation of a learner in the form of a temporal signal. Using the proposed ensemble, four affective states were estimated, namely boredom, confusion, frustration, and engagement. Dataset for Affective States in E-Environment (DAiSEE) was used to train, validate, and test the baseline model, which reported an accuracy of 86% on 4305 test samples. In the next stage, mappings between estimated affective states and learning affects (i.e. positive, negative and neutral) were established based on an adaptive mapping mechanism, to consolidate the mapping between affective states and learning affects. Live testing and survey feedback were then used to further validate, adapt and amend the feedback process. Incorporating and interpreting the estimated affective states and learning affect is imperative in providing information to both teachers and learners, and hence potentially improve the existing e-learning platforms.

Keywords: Affective states recognition · E-Learning · Adaptive Reasoning · Learning Affect.

1 Introduction

E-learning has recently risen rapidly and gained popularity due to COVID-related implications. Most universities around the world had taken to emergency

online teaching and learning during the COVID-19 pandemic, and online teaching and learning has now become a new norm. Gaining insights into learners' learning processes during these online teaching sessions is crucial for learners, teachers and curriculum developers. Peer evaluations, surveys, open-ended questions, drag-and-drop exercises, quizzes, and gamification are some of the approaches used to analyze the (learning) affect experienced by learners. Despite their widespread use, these methods do not offer teachers feedback in real-time during a teaching and learning session. Teachers in traditional classrooms judge the learning affect experienced by learners mainly based on facial expressions, body language, eye gaze, and other physical characteristics. Teachers also use query sessions to obtain feedback from learners to estimate their level of comprehension. In recent years, researchers have applied machine learning technologies to classify cues expressed by students, such as facial expressions and predict their learning affect in both classroom and online settings. Krithika et al. [21] have indicated the use of emotion and facial expressions to predict student learning affect as compared to other input modalities. However, evaluating a learner's comprehension in a real-time online setting using affective sates expressed and providing timely feedback is a complex problem, and the authors of this paper propose a baseline framework to address this issue. The goal of relevant feedback is based on reasoning between the cues exhibited by the learners and those inferred by the teachers(machine learning model in this context).

Reasoning is generally thought of as an intelligent behaviour displayed by living beings. Human beings naturally develop the ability to reason about the relationships between what they can visually witness, and thereby draw logical inferences about the real condition of the relationship that exists between the items that were being observed. The capacity to extract a meaningful pattern from a stream of data is a difficult problem for machines to achieve according to Santoro et al. in [36]. In recent years, much attention has been paid to the idea of reasoning that leads to many interesting studies in the field of image understanding, signal processing, robotics, and general artificial intelligence. Taking into account diverse observations and interest in reasoning tasks by researchers, it would be beneficial if the notion could be included in the education, particularly in e-learning for learning affect estimation and feedback generation.

James et al. and Julie et al. in [33] [32] describe learning affect as the attitudes, interests, and values that a learner exhibits and acquires at school that could have a much greater impact on his/her post-secondary life than the academic results. Reasoning about the learning affect experienced by a learner is important for many reasons, such as: (i) it allows teachers to make an inference about a learner, which might be used to design subsequent lessons [22] (ii) aids in the design of affect-responsive learning environments, which can help students control their emotions and have a more engaging learning experience [11] (iii) to recognize and respond to emotional states that are detrimental to learning, as well as produce those that are beneficial to learning, and to aid and improve learning outcomes [6] (iv) help teachers better understand learning processes by offering non-intrusive, real-time tracking of fleeting affect-cognitive interactions

that are missed by traditional self-reporting methods [6] (v) expands the scope of learning phenomena by detection of affects [6] (vi) help provide a better targeted instruction and a successful learning experience as teachers are aware of their learners' emotional traits [15].

Given the significance of learning affect, it is critical to recognize that reasoning about the learning affect of learners is an important tool in developing feedback. Feedback is an important part of the teaching and learning process as it allows learners to identify learning gaps and assess their progress [5]. It is thus envisaged that incorporating the concept of reasoning about a learners' learning affect into existing platforms and providing feedback will help aid the overall teaching and learning process.

The remainder of the paper is structured as follows: Related work on affect analysis and reasoning and its application is provided in section 2. Section 3 describes the proposed framework, which includes data and feature sampling, the training and validation experiments of the proposed model, experimental results on DAiSEE, affective states to learning affect mapping (ASLA), affective states to basic emotions, which is to validate the ASLA mapping, the live test of the proposed model. The conclusions and suggestions for future work are provided in Section 4.

2 Related work: Affect Analysis and Reasoning

2.1 Affect Analysis

Ayvaz et al.[4] presented an information system to recognize students' immediate and weighted emotional states in a virtual classroom based on their facial expressions. Happiness, surprise, fear, disgust, anger, sadness, and neutral emotions were recognized. Happiness and surprise were regarded as positive by the authors, while fear, disgust, anger, and sadness were regarded as negative. The instantaneous emotional states of the students were displayed using various emojis, allowing educators to see the general motivational levels of the virtual classroom with a motivational level chart. Accuracy of 96.38% and 98.24% was reported for the classification of emotional state using K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) respectively.

Zakka et al. [42] investigated a standardized mapping method between facial emotions and learning affects. Their research proposed a framework for estimating and classifying facial emotions based on exhibited facial expressions of anger, disgust, fear, happy, sadness, surprise and neutral. A mapping mechanism to map emotional states to learning affect experienced by a student was also presented. A CNN was trained, validated and tested using Facial Emotion Recognition-2013 dataset (FER2013), and an accuracy of 73.12% was reported on the test set. Asaju et al. [3] proposed a framework that uses a machine learning architecture for handling temporal data gathered during an online session towards estimating a student's learning affect. Emotions related to learning, i.e., boredom, frustration, engagement and confusion were estimated. On 6546 test

samples, a baseline CNN-BiLSTM model that was trained and tested using the DAiSEE dataset reported an accuracy of 88%. The estimated emotions were further categorized into affective states such as positive or negative learning affect.

Mukhopadhyay et al.[25] examined students' facial expressions to track emotional changes that take place throughout online learning. Using a CNN model, the authors classified basic emotions of anger, disgust, fear, happy, surprise, sadness and neutral. They subsequently determined the students' state of mind using estimated emotions. For emotion classification and state-of-mind identification, their model reported accuracies of 65% and 62%, respectively. To assess the levels of learning affect and engagement of students, Pise et al. [31] a Temporal Relational Network (TRN) framework which identifies the changes in facial emotions of students during an e-learning session. The use of single-scale, multi-scale TRN and an Multi-Layer Perceptron (MLP) model were explored. Samples of basic emotions of anger, disgust, fear, happy, surprise, sadness and neutral from DISFA dataset were used and accuracies of 92.7%, 89.4%, and 86.6% were reported using multiscale TRN, single-scale TRN, and MLP, respectively. Dewan et al. [12] examined how facial images could be used to estimate student engagement in an online context. An InceptionNet based framework reported accuracies of 36.5%, 47.1 %, 70.3%, and 78.3% for recognizing boredom, engagement, confusion and frustration states respectively.

Estimation of affective states using exhibited facial cues is a well researched problem and has a potential to further enhance the e-learning space. It is thus proposed to use facial cues as a modality to estimate the affective state of a learner and thus the learning affect.

2.2 Reasoning and its Applications

Li et al. [23] presented an Adaptive Hierarchical Graph Network (AHGN) based reasoning model with a semantic coherence technique for video and language inference. Their network does in-depth reasoning in three hierarchies over video frames and subtitles, with the graph reasoning structure being adaptively defined by the semantic structures of the statement. They also introduced a Semantic Coherence Learning algorithm (SCL) for the AHGN to support cross-modal and cross-level semantic coherence. Their AHGN + SCL framework reported 71.38% accuracy using VIOLIN (VideO-and-Language Inference) dataset. Zhang et al. [43] proposed a generative visual dialogue system that has an embedded adaptive reasoning module for multimodal input and reported an accuracy of 77.55%. Their module can accommodate various dialogue scenarios and selects relevant information accordingly. Their generative visual dialogue system was trained using weighted likelihood estimation (WLE). Sailer et al. [35] used an autonomous adaptive feedback mechanism based on Natural Language Processing (NLP) approaches in a simulation setting. The method was used to provide feedback on pre-service teachers' written explanations of their diagnostic reasoning for simulated students with learning disabilities. In this context, diagnostic reasoning is stated in diagnostic accuracy (i.e. whether the diagnosis is correct) and quality of

diagnostic justification (i.e., how well relevant supporting information for diagnosis is presented). In their work, the authors compared the effects of automatic adaptive NLP-based feedback against static feedback (i.e., expert solutions). The authors opined that learners' understanding of their existing state of knowledge can be aided by automatic adaptive feedback, which can indicate where and how to improve.

Reviewing the existing literature in the areas of affective states estimation, adaptive reasoning, and feedback mechanisms, it is clear that these concepts have not been fully explored in the domain of e-learning. Current work proposes a framework that captures affective states and learning affect experienced by learners based on the exhibited facial expression, and adapting generic feedback templates based on reasoning. The framework was then tested with live participants who provided their feedback of the estimates made by the model with respect to emotional state and learning affect.

3 Proposed Framework

An overview of the proposed framework is provided in Figure 1. Objectives include (i) data acquisition, (ii) feature extraction using pre-trained ResNet-152, (iii) recognition of affective states using a Bi-LSTM classifier and estimation of learning affect (ie., positive, negative and neutral) based on initial mapping formulated from literature, (iv) further validation of initial mapping based on basic emotions (i.e., anger, disgust, fear, happy, neutral, sadness, and surprise) and learning affects using samples from DISFA+ dataset, (v) evaluation of the final model using live samples and survey.

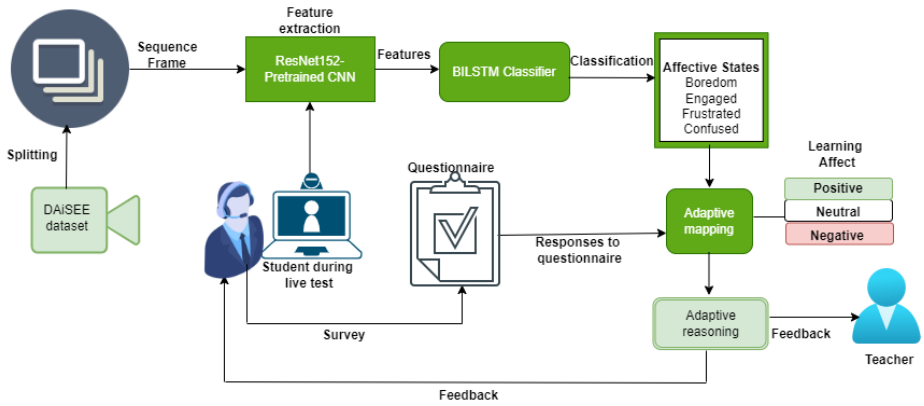

Fig. 1: The proposed framework

3.1 Data and Feature Sampling

Dataset for Affective States in E-learning Environment (DAiSEE) is an open source dataset created to study user engagement and contains 9,068 video sequences from 112 Asian subjects. Crowd annotations were provided as labels representing four affective states related to learning such as boredom, engagement, frustration and confusion. This dataset contains 10 second video samples with a resolution of 640×480 pixels/frames and a frame rate of 10 fps. A total of 2,723,882 sequences are available. Benchmark studies by [18,1] signified the use of DAiSEE in user engagement studies. [18] used DAISEE to test their user engagement prediction model(Long-term Recurrent Neural network) and reported a mean square error (MSE) of 0.0422 and an accuracy of 58.84%. On the other hand, [1] presented an end-to-end spatio-temporal hybrid architecture (ResNet + TCN) to monitor student engagement in an online classroom scenario. They reported state-of-the-art results in classifying student engagement using samples from DAiSEE dataset for model training. Their result reported was 63.9% on 1784 test videos.

Our proposed work uses DAiSEE dataset for baseline training, validation and testing owing to its success in similar studies. A total of 28,634 sequences were used with 24,329 for training and validating the model. The original 480 $\times$ 640$\times$ 3 frames were reshaped to 240 $\times$ 320 $\times$ 3 and were passed through the ResNet-152 [30]. ResNet architecture incorporates a residual learning unit, which helps improve the classification accuracy without adding to the complexity of the model [27]. An output feature vector of 2,048 is sampled at the last max-pool layer for all the 24, 329 samples resulting in a 24,329 $\times$ 2,048 size 2D vector for training and validating the model.

3.2 Training and Validating Bi-LSTM Model

The extracted features were then fed to a Bi-LSTM network, a version of the LSTM model that has two LSTMs. The first LSTM takes into account the forward structure of the input sequence, whereas the second LSTM takes into account the reverse structure. This type of LSTM can help a model learn long-term dependencies and improve its accuracy, as stated by Rahman et al. [34]. Evaluation metrics such as accuracy, precision, recall, and F-1 score were used to validate the model as stated in [19]. The Bi-LSTM model was trained using 19,638 samples and validated using 6,546 samples. A binary cross-entropy error function was used to calculate the probability of the output layer. A batch size of 128, maximum epochs of 100 with early stopping was used to avoid overfitting.

3.3 Experimental results on DAiSEE

The trained model was tested on 4,305 sequence samples from DAiSEE dataset that were not part of train and validation sets. The CNN-BiLSTM model reported an accuracy of 86% and an average F-1 score of 87%, which is a comparable performance to some other state-of-the-art methods [18,1,24,16]. Table 1

shows these comparisons. Figure 2 shows the confusion matrix of the result, while Table 2 provides a summary of the test results. The model reported a misclassification of 1.91% for boredom, 9.49% for confusion, 15.34% for engagement, and 21.38% for frustration affective states. A total of 12.03% misclassifications was reported.

Authors	Dataset	Model	Accuracy reported
[18]	DAiSEE	LRCN	58.84%
[1]	DAiSEE	ResNet + TCN	63.9%
[24]	DAiSEE	SENET +LSTM+GALN	58.84%
[16]	DAiSEE	STGAT	
Current Study	DAiSEE	CNN-BiLSTM	**86%**

Table 1: Comparison of result with state-of-the art methods

Emotions	Number of samples	Precision	Recall	F1-Score
Boredom	782	0.98	0.91	0.94
Confusion	758	0.89	0.81	0.85
Engagement	1362	0.84	0.79	0.81
Frustrated	1403	0.81	0.94	0.87
Average		**0.88**	**0.86**	**0.87**

Table 2: Summary of the test reported on DAiSEE dataset

3.4 Affective States to Learning Affects(ASLA) : Initial Mappings

According to Sheng et al.[7] affective states experienced by learners can be mapped onto positive, negative and neutral learning affects. A number of studies have been carried out in this regard. Daschmann et al. [10] reported that students who experience high levels of boredom are more likely to have negative outcomes such as poor grades, absentism, and eventually dropping out from school, indicating a negative affect on learning. On similar lines, D'Mello et al. [13] opined that students become bored when the lecture does not interest them or when they are unable to handle any of the tasks that are part of the learning process, resulting in a poor learning affect. In their subsequent study, D'Mello et al. [14] opined that confusion is often caused by a lack of understanding of the subject, especially when students face a challenge during their learning activities or are unclear about the knowledge being disseminated, concluding that confusion is a sign of negative learning affect. When students are confused, especially

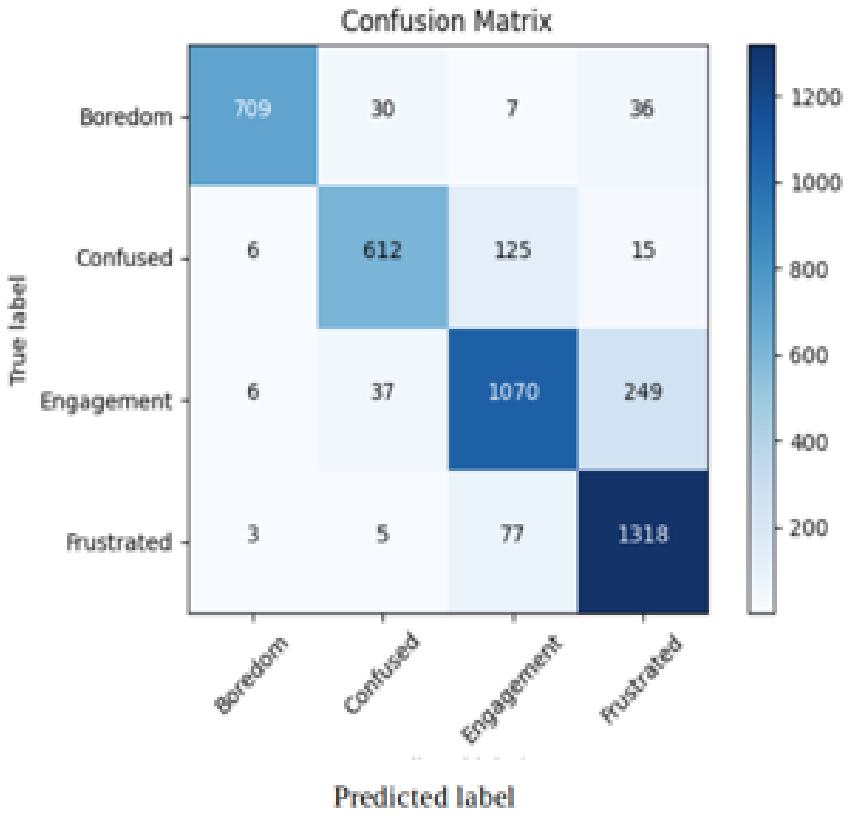

Fig. 2: Confusion matrix showing the test experiment's results

when there is a need to understand or comprehend a lecture, they can become frustrated and make repeated mistakes [13,41]. As a result, frustration was also associated with a negative learning affect. The authors of [9,28] associated an intense feeling of flow-like engagement, especially when the learning outcomes are clear and the lecture content is well understood, with a positive learning affect. Table 3 illustrates mapping between affective states and learning affect based on the works carried out in this space.

3.5 Affective States to Basic Emotion : Validating ASLA Mapping

Experiments to consolidate the mapping between affective states and learning affects were carried out by investigating the mapping between affective states and emotions. Previous work on mapping between basic emotions and learning affects (as shown in Table 4) was used as a baseline for the verification and validation of the mappings between affective states and learning affects.

Affective States	Literature	Mapped to
Boredom	[10,2,3],	Negative
Confusion	[14,2,3],	Negative
Frustration	[13,41,9,8,2,3]	Negative
Engagement	[9,39,17,26,2,3]	Positive

Table 3: Mapping of affective states to learning affects based on work by Dachmnn et al.[10], Asaju et al.[3,2], D'Mello et al.[14], Stein et al.[41], Csikszentmihalyi et al.[9,8], Shernoff et al.[39], Guo et al.[17] and Nakamura et al.[26]

Emotions	[38]	[20]	[29]	[42]	[2,40]
Anger	Negative	Negative	Positive	Negative	Negative
Disgust	-	Negative	Negative	Negative	Negative
Fear	Positive	Positive	-	Positive	Positive
Happy	-	Positive	Positive	Positive	Positive
Sadness	Negative	Negative	-	Negative	Negative
Surprise	Positive	Positive	Positive	Positive	Positive
Neutral	-	-	Positive	Positive	Neutral

Table 4: Mapping of facial emotion to learning affect by Sathik et al. [38], Kapoor et al.[20], Pan et al. [29], Zakka et al.[42] and Asaju et al.[2,40]

The objective behind this experiment was to determine whether the estimated affective states of engagement, frustration, boredom, and confusion are mapped to the relevant basic emotions (e.g., anger, disgust, fear, happy, surprise, sadness, and neutral). To achieve this objective, authors tested the DAiSEE trained model on a set of 4755 unlabelled samples from the DISFA + facial expression dataset. The results gathered from using samples labelled with emotion states to estimate affective states are provided in Table 5 and are discussed as follows:

Samples labelled as depicting "happy" and "surprise" emotions (from DISFA+) were classified as "engagement" affective states by our trained model. Work by [20,42,37,29] maps "happy" and "surprise" emotions to a positive learning affect, consolidating our mapping of "engagement" affective state to a positive learning affect. This is in line with the mappings provided in [8,26,9,39,17]. Generally, it is believed that when students concentrate or are engaged during a learning session, it shows that they have some level of understanding of the lecture. This can result in positive learning affect and can eventually cause a better performance in that subject. This study opines that the expressions of happy, surprise, and /or affective states of engagement shown by students during learning are highly likely to result in a positive learning affect.

Samples labelled as depicting "sadness" emotion (from DISFA+) were estimated as affective state of "confusion" by our trained model. The works by [38,20,42], mapped sadness emotion to a negative learning affect, consolidating our mapping of "confusion" affective state to a negative learning affect. This is inline with the mappings provided by D'Mello et al. [14] . Traditionally, sadness

emotions depict a level of uncertainty or unhappiness or dis-satisfaction about a situation, and so exhibit affective states of confusion. This could mean that they may not have a good understanding of the lecture and, therefore, the likelihood of a negative learning affect is very high.

Samples labelled as depicting "fear" emotion (from DISFA +) were estimated as "engagement" affective states by our trained model. Fear was considered positive learning affect by the authors in [38,20,42], consolidating our mapping of "engagement" affective states as a positive learning affect. This corresponds with the mappings of works in [9,8,26,39,17]. Traditionally, fear could be considered as depicting a negative condition, but through a literature survey and results from experiments and the survey that was carried out using students, it was suggested that fear that was detected during learning means a positive learning affect.

Samples labelled as depicting "disgust" emotion (from DISFA+) were estimated as "boredom" affective state by our trained model. Disgust emotions were considered negative emotions by the authors in [20,29,3], consolidating our mapping of "boredom" affective state to a negative learning affect. This corresponds to the mappings provided in [10,13]. This study agrees with the mapping of disgust emotion and the affective state of boredom as a negative learning affect, as provided by the literature and the proposed model. This study infers that when students become bored during a learning session, they can depict disgust emotions, and this could result in a likelihood of a negative learning affect.

Samples labelled as depicting "anger" emotion (DISFA +) were estimated as "frustration" affective state by our trained model. The authors of [42,38,20], mapped "anger" emotion to a negative learning affect, consolidating our mapping of "frustration" affective state to a negative learning affect. This is in line with the mappings provided in [9,26]. Generally, emotions of "anger" and affective states of "frustration" are negative signs in any scenario, and this study opines that the expressions of "anger" and affective states of "frustration" shown by students during learning are highly likely to result in a negative learning affect.

DISFA+ Labels	Sample size	Estimated Affective states by proposed model	Mapped Learning Affect
Anger	822	Frustration	Negative
Disgust	782	Boredom	Negative
Fear	758	Engagement	Positive
Happy	851	Engagement	Positive
Sadness	751	Confusion	Negative
Surprise	791	Engagement	Positive
Total	4755		

Table 5: Validating mapping between Affective States and Learning Affects using existing mapping between emotions and learning affects

3.6 Live Testing of the Proposed Model

It is imperative to validate models through live testing to understand the effectiveness and pitfalls of a proposed framework. The authors propose to evaluate and examine the workings of the model by conducting experiments in semi-controlled laboratory settings, which involve computer science students at the institution where this research is carried out. As this involves human participants, an application for ethical clearance was sent to the University's Human Research Ethics Committee (non-medical) for approval. Survey questionnaires, participation and consent forms used as part of our application are provided in Appendix. Ethics clearance No: H21/05/02.

The proposed system was tested by 22 students from the Computer Science department. After going through the participation and consent forms, participants were required to face a video recorder while watching a 10-minute lecture on a given topic. The recorded video sequence depicting the facial cues exhibited by the participant is then fed to the proposed model to estimate an affective state. The affective states that were depicted more in a sample were considered to be the highest depicted affective states in a sample. The second highest depicted affective states were also recorded. This study considered the highest occurring affective states for analysis. After this session, questionnaires were administered to participants to compare the affective states as experienced by the participants themselves (in terms of their basic emotions) with those estimated by the model. This is to investigate the relationship between affective states and basic emotions and to validate the previous mapping of affective states to basic emotions. The obtained results are illustrated in Table 6.

The estimates from our trained model indicated that some affective states were depicted more frequently than others. It was observed that a majority of the participants/learners(59.09%) expressed engagement affective state as the highest occurring affective state, followed by frustration (13.64%) and confusion (27.27%). On the other hand, 40.91% of the respondents estimated happy emotions as their feelings during the lecture, and an additional 9.09% of the participants reported surprise. This totals to a 50% towards the affective state of engagement. 13.64% of the participants reported sadness as their experienced emotion as they did not understand what was being taught. This was inline with the number of participants who were labeled as exhibiting an affective state of frustration, which is a negative learning affect. 36.36% of participants reported neutral expression.

We summarize the survey results to validate our mappings as validated through various experiments and literature in this study in Table 7.

Drawing on several studies and experiments carried out, the current study confirms that when students are experiencing learning, they can experience a variety of affective states that can provide feedback to both teachers and students. Furthermore, research reveals that affects expressed and learning affects are intertwined.

Participant	Estimates affective states with the highest probability	Estimated affective state with second highest probability	Participant's estimates in terms of basic emotions
1	Engagement	Engagement	Happy
2	Frustration	Confusion	Neutral
3	Confusion	Boredom	Sadness
4	Engagement	Engagement	Happy
5	Confusion	Frustration	Sadness
6	Confusion	Frustration	sadness
7	Engagement	Engagement	Happy
8	Engagement	Confusion	Neutral
9	Confusion	Engagement	Neutral
10	Engagement	Confusion	Neutral
11	Engagement	Engagement	Happy
12	Frustration	Boredom	Neutral
13	Frustration	Boredom	Neutral
14	Engagement	Engagement	Happy
15	Engagement	Confusion	Happy
16	Confusion	Engagement	Neutral
17	Engagement	Engagement	Surprise
18	Engagement	Confusion	Happy
19	Confusion	Engagement	Neutral
20	Engagement	Frustration	Surprise
21	Engagement	Engagement	Happy
22	Engagement	Engagement	Happy

Table 6: Model and participants' estimates during live test.

Affective state/Emotions	**Learning Affect**
Sadness, Frustration, Confusion	Negative
Happy, Surprise, Engagement	Positive
Neutral	Neutral

Table 7: Adaptive reasoning summary

4 Conclusion and Future Work

This paper proposed an adaptive reasoning framework utilizing a machine learning approach. The framework was able to classify four affective states of boredom, frustration, engagement, and confusion. Using DAiSEE dataset, an accuracy of 86% was reported. Various deep learning approaches have used DAiSEE dataset, such as work in [18,1,24,16] etc, the approach used in this work has achieved a comparable performance to other state-of-the-art methods. Estimated affective states were mapped into learning affect of positive, negative and neutral categories, based on a survey of relevant literature. The study further tested the proposed framework with emotion dataset; DISFA+. The purpose was to estimate the labels for samples from DISFA+ dataset that had been labelled for

basic emotions, and to verify if the estimated affective states mapped into the relevant basic emotions, and thus to verify the initial mapping of affective states. Live testing experiments and survey feedback were also carried out. The general findings of the study revealed that there is a correlation between facial emotions and learning affect. The work's findings show that using estimation of emotion/affective states and mapping to learning affect for analysis provides a better understanding of the student's learning, particularly in e-learning systems where physical contact is not possible. The proposed system is intended to serve as a first step toward the development of complete e-learning platforms with adaptive feedback incorporated. The authors also would like to explore the use of eye gaze, head pose, physiological signals, body posture, and other input signals along with feedback generation based on reasoning.

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Consent Form

Title of project: Adaptive reasoning : An affect Related feedback approach for enhanced e-learning

Name of researcher : **xxxxxxxxxxx**

I, …….., agree to participate in this research project. The research has been explained to me and I understand what my participation will involve. I agree to the following:

(Please circle the relevant options below).

I agree that my participation will remain anonymous YES NO

I agree that the researcher may use anonymous quotes in his / her research report YES NO

I agree that my participation in the data collection process may be video recorded YES NO

…..……(signature).....(name of participant)

…… (date)

…… (signature).....(name of person seeking consent) (date)

Adaptive reasoning : An affect Related feedback approach for enhanced e-learning

Questionnaire

This study intends to improve upon the current set-up in the e-learning environment by the analysis of facial emotion recognition, learning affect estimations and reasoning regarding a student's understanding. As part of the studies, I am investigating the emotional changes in students during an E-learning class under the supervision of XXXXXX. In the study after conducting a live test, which includes detection and classification of the emotions, will map the emotions that is being expressed into positive, negative, and neutral learning affects. The intention is to apply this system to the e-learning domain in order to estimate the learning affects of students. As part of the work, the researcher would like you to answer the following questions in attempt to carry out a survey to get a further understanding of the accuracy of the system that was trained, validated, tested and which a live testing is being conducted.

Please answer the following questions which is based on the tutoring video, the emotion and the learning affect estimation. Please note that your answers will be kept anonymous.

Thank you.

Questions	**Answers**							
1. How do you feel about the lecture? Please respond to this question by picking from the list of emotions given.	Anger	Disgust	Fear	Happy	Neutral	Sadness	Surprise	
2. What category of learning affect will you classify your learning outcomes from the lecture?	Positive	Negative	Neutral					
3. Give a rating of the lecture you have watched. For example, how well it was presented.	30%	40%	50%	60%	70%	80%	90%	100%
4. Is facial emotion expression a good method to evaluate a student in an online learning?	YES	NO	I don't know	Comment				
5. Do you think it is a good measure to improve upon the e-learning environment by estimating the learning affects of students through their facial emotion expression?	YES	NO	I don't know	Comment				
6. Can this method provide an insight to learner's comprehension level.	YES	NO	I don't know	Comment				
7. Can this study help teachers to adequately evaluate the students during online teaching sessions?	YES	NO	I don't know	Comment				
8.Suggest other criteria that could be used to estimate learning affect in online learning platform.								
9. Can this system be improved upon in the future?	YES	NO	I don't Know	Comments				
10. If yes for question(9)Suggest ways by which it can be improved upon.	It can be improved upon by……							

Fig. 4: Consent form and questionnaire